DAEN/ISEN 427 Homework 4 [Practice questions for exam #2]

Due date: Tue Oct 28, 2025 (before class)

Rubric: These questions are provided solely for reference and practice in preparation for the 2nd exam. They will not be graded. Any submission that demonstrates any reasonable Bayesian analysis and a minimum level of effort will receive one point for participation.

1. Compare priors Beta$(0.5, 0.5)$, Beta$(1, 1)$, and Beta$(1, 4)$. How do the priors differ in terms of shape?

   Answer: The probability density function (pdf) of a Beta distribution is given by:

   $$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 < x < 1,$$

   where $B(\alpha, \beta)$ is the Beta function.

   - **Beta(0.5, 0.5):** This distribution is *U-shaped*, with most of its mass near 0 and 1. It strongly favors extreme probabilities (close to 0 or 1). This corresponds to a prior belief that the probability parameter is very likely to be near either 0 or 1, but not in between.
   - **Beta(1, 1):** This is the *uniform distribution* on $[0, 1]$. It represents a completely *non-informative prior*, giving equal weight to all probabilities between 0 and 1.
   - **Beta(1, 4):** This distribution is *skewed to the right*, with most of its mass near 0. It implies a prior belief that smaller probabilities (near 0) are more likely, while large probabilities (near 1) are less likely.

   Graphically: (*i*) Beta$(0.5, 0.5)$: U-shaped, (*ii*) Beta$(1, 1)$: Flat line. (*iii*) Beta$(1, 4)$: Right-skewed, concentrated near 0.

2. In the following definition of a probabilistic model, identify the prior, the likelihood, and the posterior:

   $$Y \sim \mathcal{N}(\mu, \sigma) \qquad \mu \sim \mathcal{N}(0, 1) \qquad \sigma \sim \mathcal{HN}(1)$$

   $\mathcal{HN}$: the half-normal distribution is a fold at the mean of an ordinary normal distribution with mean zero.

   Hint: $f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

   Answer:

   Explanation (in words):

   (a) The priors define beliefs about parameters before observing data.

   (b) The likelihood models how data are generated given parameters.

   (c) The posterior updates the priors using observed data through Bayes' rule.

   - **Prior:** The priors are placed on the unknown parameters μ and σ:

     $$\mu \sim \mathcal{N}(0, 1), \quad \sigma \sim \mathcal{HN}(1)$$

     where $\mathcal{HN}(1)$ denotes a Half-Normal distribution with scale parameter 1.
   - **Likelihood:** Given parameters μ and σ, the data Y are assumed to be normally distributed:

     $$Y \mid \mu, \sigma \sim \mathcal{N}(\mu, \sigma)$$

     The likelihood function for observed data y is therefore:

     $$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$
   - **Posterior:** By Bayes' theorem, the posterior distribution combines the likelihood and the priors:

     $$p(\mu, \sigma \mid y) \propto p(y \mid \mu, \sigma)\, p(\mu)\, p(\sigma)$$

     Substituting the expressions:

     $$p(\mu, \sigma \mid y) \propto \frac{1}{\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) \exp\left(-\frac{\mu^2}{2}\right) \exp\left(-\frac{\sigma^2}{2}\right), \quad \sigma > 0$$

3. In the previous model, how many parameters will the posterior have? Compare your answer with that from the model in the coin-flipping problem in equation:

$$\theta \sim \text{Beta}(\alpha, \beta) \qquad Y \sim \text{Bin}(n = 1, p = \theta)$$

Answer:

Normal model: For $Y \sim \mathcal{N}(\mu, \sigma)$, $\mu \sim \mathcal{N}(0, 1)$, $\sigma \sim \mathcal{HN}(1)$, the unknowns are μ and σ. Hence the posterior is a *joint* distribution $p(\mu, \sigma \mid y) \propto p(y \mid \mu, \sigma)\, p(\mu)\, p(\sigma)$, which has **two parameters** (dimension 2): μ and σ.

Coin-flipping model: For $\theta \sim \text{Beta}(\alpha, \beta)$, $Y \sim \text{Bin}(n = 1, p = \theta)$, there is a single unknown θ. The posterior is $\theta \mid y \sim \text{Beta}\big(\alpha + y,\ \beta + 1 - y\big)$, so the posterior has **one parameter** (dimension 1): θ.

Comparison: The posterior dimension equals the number of unknown model parameters: 2 for the Normal model (μ, σ) vs. 1 for the Bernoulli/Beta model (θ).

4. Suppose that we have two coins; when we toss the first coin, half of the time it lands tails and half of the time on heads. The other coin is a loaded coin that always lands on heads. If we choose one of the coins at random and observe a head, what is the probability that this coin is the loaded one?

Answer: Let:

$$C_1 = \text{fair coin}, \quad C_2 = \text{loaded coin}$$

$$P(C_1) = P(C_2) = 0.5$$

The likelihoods are:

$$P(H \mid C_1) = 0.5, \quad P(H \mid C_2) = 1$$

We observe a head (H) and we want:

$$P(C_2 \mid H)$$

By Bayes' theorem:

$$P(C_2 \mid H) = \frac{P(H \mid C_2)P(C_2)}{P(H \mid C_1)P(C_1) + P(H \mid C_2)P(C_2)}$$

Substituting values:

$$P(C_2 \mid H) = \frac{(1)(0.5)}{(0.5)(0.5) + (1)(0.5)} = \frac{0.5}{0.25 + 0.5} = \frac{0.5}{0.75} = \frac{2}{3}$$

$$\boxed{P(C_2 \mid H) = \tfrac{2}{3}}$$

Interpretation: Given that a head was observed, there is a $\frac{2}{3}$ probability that the coin chosen was the loaded one.

5. Match the following functions that play a role in Bayesian modeling with the descriptions.

   Functions:

   (a) Distribution of the posterior mean estimate
   (b) Prior distribution
   (c) Likelihood function
   (d) Posterior distribution
   (e) Measurement distribution

Descriptions:

(a) Is the result of inference on an individual trial
(b) Describes how potentially noisy observations are generated
(c) Can be directly compared to human responses in a psychophysical experiment
(d) Is often modeled as a Gaussian function centered at the measurement
(e) May reflect statistics in the natural world

Answer:

$$1 \to 3, \quad 2 \to 5, \quad 3 \to 2, \quad 4 \to 1, \quad 5 \to 4$$

Function	**Matching Description**
1. Distribution of the posterior mean estim.	3. Can be directly compared to human responses in a psychophysical experiment
2. Prior distribution	5. May reflect statistics in the natural world
3. Likelihood function	2. Describes how potentially noisy observations are generated
4. Posterior distribution	1. Is the result of inference on an individual trial
5. Measurement distribution	4. Is often modeled as a Gaussian function centered at the measurement

6. You need to estimate the weights of blue whales, humans, and mice. You assume the weights are normally distributed, and you set the same prior $\mathcal{HN}(200 \text{ kg})$ for the variance (positive scatter parameter). What type of prior is this for adult blue whales? Strongly informative, weakly informative, or non-informative? What about for mice and for humans? How does informativeness of the prior correspond to our real world intuitions about these animals?

   Answer: Let $\sigma \sim \mathcal{HN}(200 \text{ kg})$. A Half-Normal with scale 200 puts most mass on σ below a few hundred kilograms.

   - **Blue whales:** adult weights are on the order of 10^5 kg and realistic population scatter is in the *thousands* of kg. A prior concentrating on $\sigma \lesssim$ a few 100 kg is therefore *strongly informative* (and overly tight) in this context.
   - **Humans:** adult human weights are ~ 70 kg with σ roughly 10–15 kg. Compared with that scale, $\mathcal{HN}(200 \text{ kg})$ is very broad. Thus it is *weakly informative* (close to non-informative) for humans.
   - **Mice:** adult mice weigh ~ 25 g $(0.025$ kg$)$ with σ only a few grams (~ 0.003 kg). Relative to this scale, $\mathcal{HN}(200 \text{ kg})$ is effectively flat over any plausible region, so it is *non-informative*.

   Takeaway: Prior *informativeness is relative to the problem's scale*. The same numeric prior can be overly informative for large organisms (blue whales) yet weak or effectively non-informative for smaller ones (humans, mice). Prior predictive checks should be used to assess this mismatch before fitting.

7. A student says, "to find the probability distribution of an observer's estimate $\hat{s}$ for a given stimulus s, I can just multiply the measurement distribution $p(x \mid s = \mu)$, where μ is the prior mean, by the prior $p(s)$."

   In other words, the student writes:

   $$\text{(wrong)} \quad p(\hat{s} \mid s) = p(x \mid s = \mu)\, p(s).$$

   (a) This formula gives a result that might look reasonable but it is conceptually wrong. Explain why.
   Answer:

   - $p(x \mid s)$: how the *measurement* x depends on the true stimulus s (this is the sensory noise model).
   - $p(s)$: the *prior belief* about which stimuli are likely, before seeing any data.
   - $p(\hat{s} \mid s)$: how the observer's *estimate* $\hat{s}$ varies when the true stimulus is s.

   What the student did wrong: he/she replaced s with the prior mean μ in the measurement model, using $p(x \mid s = \mu)$. That means he/she is using the measurement distribution for a *fixed average stimulus*, not for the actual stimulus shown.

Then he/she multiplied it by the prior $p(s)$, which describes beliefs *before* seeing any data. This mixes up two different things: (*i*) the **data generation process** $p(x \mid s)$ and (*ii*) the **prior belief** $p(s)$. These belong to different steps in the probabilistic model, so multiplying them is not correct.

The correct way to find how the estimate $\hat{s}$ varies for a given true stimulus s is:

$$p(\hat{s} \mid s) = \int p(\hat{s} \mid x)\, p(x \mid s)\, dx.$$

This means:

i. The true stimulus s produces a noisy measurement x.
ii. The observer uses that measurement x to make an estimate $\hat{s}$.
iii. You average (integrate) over all possible noisy measurements.

In short, the formula is wrong because it mixes up the roles of the prior and the measurement model. The prior $p(s)$ should only appear in the observer's inference step

$$p(s \mid x) \propto p(x \mid s)\, p(s),$$

not in $p(\hat{s} \mid s)$.

To find $p(\hat{s} \mid s)$, we must integrate over possible noisy measurements:

$$p(\hat{s} \mid s) = \int p(\hat{s} \mid x)\, p(x \mid s)\, dx.$$

	Correct relationship	**Student idea**
Goal	Distribution of $\hat{s}$ given true s	Multiply measurement and prior directly
Formula	$p(\hat{s} \mid s) = \int p(\hat{s} \mid x)\, p(x \mid s)\, dx$	$p(\hat{s} \mid s) = p(x \mid s = \mu)\, p(s)$
Mistake	Uses correct chain $s \to x \to \hat{s}$	Uses prior wrongly; ignores dependence on s

(b) Write down the *correct expression* for the response distribution $p(\hat{s} \mid s)$ in terms of the measurement distribution $p(x \mid s)$ and the observer's decision rule $p(\hat{s} \mid x)$. Briefly explain what each term represents.
Answer:

$$p(\hat{s} \mid s) = \int p(\hat{s} \mid x)\, p(x \mid s)\, dx.$$

- $p(x \mid s)$: the *measurement distribution*, describing how a noisy internal measurement x is generated from the true stimulus s.
- $p(\hat{s} \mid x)$: the *observer's decision rule* or *response model*, describing how the observer converts a measurement x into an estimate or response $\hat{s}$.
- The integral sums (or averages) over all possible measurements x, giving the overall probability of producing a response $\hat{s}$ when the true stimulus is s.

(c) Sketch or describe how the *correct* response distribution $p(\hat{s} \mid s)$ would change if the measurement noise increased. Hint: think about how $p(x \mid s)$ becomes wider and how that affects the spread of responses.
Answer:

Idea: $p(\hat{s} \mid s) = \int p(\hat{s} \mid x)\, p(x \mid s)\, dx \quad \Rightarrow \quad$ wider $p(x \mid s)$ makes $p(\hat{s} \mid s)$ broader (and lower-peaked).

Description: As the *measurement noise* increases, the measurement distribution $p(x \mid s)$ becomes wider. Because $p(\hat{s} \mid s)$ averages over all x via the integral above, that extra spread in x is *propagated* through the decision rule $p(\hat{s} \mid x)$, making the response distribution $p(\hat{s} \mid s)$ more dispersed (broader) and less sharply peaked.

- **If the decision rule is roughly deterministic and unbiased** (e.g., $\hat{s} \approx x$ or a near-identity mapping), then $p(\hat{s} \mid s)$ simply *widens* around s; the mean stays near s, variance increases.
- **If the rule is Bayesian with a prior** $p(s)$ (e.g., MAP/posterior mean), then increasing measurement noise not only broadens variability but also *pulls the responses toward the prior mean* μ (regression-to-the-mean). In the extreme-noise limit, responses cluster near μ (strong bias).

How to sketch: Draw the original $p(\hat{s} \mid s)$ as a bell-shaped curve centered near the true s. Then, for higher noise, redraw it as a *flatter and wider* curve. If a strong prior is used, also shift the peak slightly toward μ.

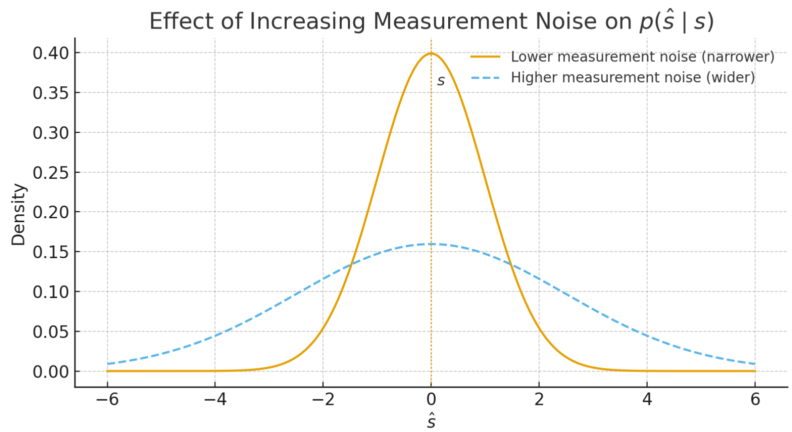

Figure 1: As measurement noise increases, $p(\hat{s} \mid s)$ becomes wider and less peaked.

8. In the *Bayesian perception framework*, an observer tries to estimate the stimulus s based on a noisy measurement m.

- $p(m \mid s)$: *Measurement distribution* — probability of observing measurement m given stimulus s.
- $p(s)$: *Prior* — how likely different stimuli are before seeing the data.
- $p(s \mid m)$: *Posterior* — probability of each stimulus given the measurement.
- $p(m)$: *Evidence (marginal likelihood)* — overall probability of the measurement.
- $p(r \mid s)$: *Response distribution* — probability that the observer gives response r for stimulus s.

These quantities are related through Bayes' rule:

$$p(s \mid m) = \frac{p(m \mid s)\, p(s)}{p(m)}, \quad p(r \mid s) = \int p(r \mid m)\, p(m \mid s)\, dm.$$

The goal is to understand how the following concepts are related: (*i*) likelihood function, (*ii*) measurement distribution, (*iii*) posterior, (*iv*) prior, and (*v*) the observer's response or estimate.

True or False? If false, please explain why.

(a) The likelihood function is always equal to the measurement distribution.
Answer: False. Although the likelihood function and the measurement distribution share the same mathematical form $p(m \mid s)$, they represent different perspectives. The measurement distribution is a function of the measurement m given a fixed stimulus s, while the likelihood is interpreted as a function of s given a fixed measurement m. Thus, they are conceptually different.

(b) The stimulus value that maximizes the posterior probability is the same as the measurement value.
Answer: False. The stimulus value that maximizes the posterior $p(s \mid m)$ is the *maximum a posteriori* (MAP) estimate. It equals the measurement value only if the prior is uniform and the likelihood is symmetric. Otherwise, the MAP estimate is shifted toward regions with higher prior probability.

(c) The response distribution for a given stimulus can be found by multiplying the measurement distribution by the prior.
Answer: False. The response distribution $p(r \mid s)$ is obtained by integrating over measurements:

$$p(r \mid s) = \int p(r \mid m)\, p(m \mid s)\, dm.$$

Multiplying the measurement distribution $p(m \mid s)$ by the prior $p(s)$ gives the joint distribution $p(m, s)$, not the response distribution.

(d) The response distribution for a given stimulus is always Gaussian.
Answer: False. Even if the measurement noise is Gaussian, nonlinear transformations in the observer's decision process can make the response distribution non-Gaussian. Thus, it is not necessarily Gaussian.

(e) If a Bayesian observer reports one stimulus value more often than another, it means the prior probability of that value is higher.
Answer: False. The observer's responses depend on both the prior and the likelihood. A stimulus may be reported more frequently due to measurement noise, decision boundaries, or asymmetries in the likelihood, not just a higher prior. Therefore, frequent reporting does not necessarily imply a higher prior probability.

9. In Bayesian inference, we often want to compute the posterior distribution

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)},$$

where θ are model parameters and D is observed data. In most realistic problems, computing the denominator

$$p(D) = \int p(D \mid \theta)\, p(\theta)\, d\theta$$

is computationally heavy (or intractable because it involves integrating over a high-dimensional parameter space).

(a) Explain why the normalization constant $p(D)$ is difficult to compute.
Answer: The normalization constant $p(D)$, also known as the *marginal likelihood* or *evidence*, is difficult to compute in high-dimensional models because it requires integrating over all possible parameter values θ:

$$p(D) = \int p(D \mid \theta)\, p(\theta)\, d\theta.$$

This computation becomes challenging for several reasons:

i. **High-dimensional integration:** As the number of parameters (the dimension of θ) increases, the integral must be evaluated over an exponentially growing space. This is known as the *curse of dimensionality*, and it makes numerical integration or sampling methods computationally expensive.

ii. **Complex likelihood surfaces:** The likelihood function $p(D \mid \theta)$ may have multiple modes or highly irregular shapes, making it difficult for numerical methods to capture all regions that contribute significantly to the integral.

iii. **Lack of closed-form solutions:** In most realistic models, the product $p(D \mid \theta)p(\theta)$ does not yield an analytically integrable expression, so $p(D)$ cannot be computed exactly.

iv. **Computational cost of evaluating the likelihood:** Each evaluation of $p(D \mid \theta)$ can be expensive for large datasets or complex models, and since the integral requires many such evaluations, the total computation becomes infeasible.

In summary, computing $p(D)$ is difficult because it involves a high-dimensional, non-analytic integral over the parameter space that is computationally expensive to approximate accurately.

(b) Describe one computational method that can be used to approximate the posterior $p(\theta \mid D)$ without evaluating $p(D)$ directly. Give an example of how this method works in principle.
Answer: One computational method to approximate the posterior $p(\theta \mid D)$ without evaluating $p(D)$ directly is Markov Chain Monte Carlo (MCMC).

MCMC methods generate samples from the posterior distribution $p(\theta \mid D)$ by constructing a Markov chain whose stationary distribution is the desired posterior.

This approach avoids computing the normalization constant $p(D)$, since sampling only requires evaluating the unnormalized posterior $p(D \mid \theta)p(\theta)$.

Example, Metropolis–Hastings algorithm:

i. Start from an initial parameter value $\theta^{(0)}$.

ii. Propose a new value θ' from a proposal distribution $q(\theta' \mid \theta^{(t)})$.

iii. Compute the acceptance ratio

$$r = \frac{p(D \mid \theta')\, p(\theta')\, q(\theta^{(t)} \mid \theta')}{p(D \mid \theta^{(t)})\, p(\theta^{(t)})\, q(\theta' \mid \theta^{(t)})}.$$

iv. Accept the new sample with probability $\alpha = \min(1, r)$; otherwise, keep the current sample.

v. Repeat to generate a sequence of samples $\{\theta^{(t)}\}$.

After a sufficient number of iterations (after the chain has *converged*), the samples $\{\theta^{(t)}\}$ are approximately drawn from the posterior distribution $p(\theta \mid D)$. These samples can then be used to estimate expectations or credible intervals without ever computing the normalization constant $p(D)$.

(c) Suppose we use a *Metropolis–Hastings* Markov Chain Monte Carlo (MCMC) algorithm to sample from $p(\theta \mid D)$. Briefly describe how a proposed sample θ' is accepted or rejected.

Answer: In the *Metropolis–Hastings* algorithm, a proposed sample θ' is accepted or rejected based on an *acceptance probability* that compares how well θ' explains the data relative to the current state $\theta^{(t)}$.

Given the current sample $\theta^{(t)}$, we:

i. Propose a new sample θ' from a proposal distribution $q(\theta' \mid \theta^{(t)})$.

ii. Compute the acceptance ratio:

$$r = \frac{p(D \mid \theta')\, p(\theta')\, q(\theta^{(t)} \mid \theta')}{p(D \mid \theta^{(t)})\, p(\theta^{(t)})\, q(\theta' \mid \theta^{(t)})}.$$

iii. Accept the proposed sample with probability

$$\alpha = \min(1, r).$$

If the proposal is accepted, set $\theta^{(t+1)} = \theta'$; otherwise, keep the current value, $\theta^{(t+1)} = \theta^{(t)}$.

This rule ensures that the Markov chain has the desired stationary distribution $p(\theta \mid D)$, allowing us to sample from the posterior without computing the normalization constant $p(D)$.

(d) Discuss one advantage and one disadvantage of using MCMC.

Answer: **Advantage:** MCMC methods, such as the Metropolis–Hastings algorithm, can approximate complex posterior distributions $p(\theta \mid D)$ even when they have no closed-form expression or exist in high-dimensional spaces. They can capture multimodal, skewed, or otherwise complicated distributions where simpler approximation methods (e.g., variational inference) may fail.

Disadvantage: MCMC methods can be computationally expensive and slow to converge, especially in high-dimensional models. The samples in the Markov chain are correlated, meaning that a large number of iterations (and careful tuning of the proposal distribution) are often required to obtain accurate estimates, making MCMC less practical for very large datasets or complex models.

10. Suppose we are estimating a parameter θ that represents the mean of a Gaussian process. We observe one noisy measurement $x = 5.2$ drawn from

$$p(x \mid \theta) = \mathcal{N}(x; \theta, \sigma^2), \quad \text{where } \sigma^2 = 1.$$

We assume a Gaussian prior on θ:

$$p(\theta) = \mathcal{N}(\theta; 0, \tau^2), \quad \text{where } \tau^2 = 4.$$

Hints:

- For a Gaussian likelihood and prior, the posterior is also Gaussian with:

$$\mu_{\text{post}} = \frac{\frac{x}{\sigma^2} + \frac{0}{\tau^2}}{\frac{1}{\sigma^2} + \frac{1}{\tau^2}}, \quad \sigma^2_{\text{post}} = \frac{1}{\frac{1}{\sigma^2} + \frac{1}{\tau^2}}$$

- The unnormalized posterior is proportional to $\exp\left[-\frac{1}{2\sigma^2}(x-\theta)^2 - \frac{1}{2\tau^2}\theta^2\right]$
- $r = e^{\log r}$

(a) Write down the unnormalized posterior distribution $p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$.
Answer: **Step 1: Likelihood**
We are told that the data x is drawn from a Gaussian distribution

$$p(x \mid \theta) = \mathcal{N}(x; \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right).$$

Since we only care about the posterior up to a proportionality constant, we can ignore normalization constants and write

$$p(x \mid \theta) \propto \exp\left(-\frac{1}{2\sigma^2}(x-\theta)^2\right).$$

Substituting $\sigma^2 = 1$ gives

$$p(x \mid \theta) \propto \exp\left(-\frac{1}{2}(x-\theta)^2\right).$$

Step 2: Prior
The prior for θ is Gaussian:

$$p(\theta) = \mathcal{N}(\theta; 0, \tau^2) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{\theta^2}{2\tau^2}\right).$$

Ignoring constants again, we write

$$p(\theta) \propto \exp\left(-\frac{\theta^2}{2\tau^2}\right).$$

Substituting $\tau^2 = 4$ gives

$$p(\theta) \propto \exp\left(-\frac{1}{2}\frac{\theta^2}{4}\right) = \exp\left(-\frac{\theta^2}{8}\right).$$

Step 3: Posterior (unnormalized)
By Bayes' rule,

$$p(\theta \mid x) \propto p(x \mid \theta)\, p(\theta).$$

Substituting the two exponential expressions:

$$p(\theta \mid x) \propto \exp\left(-\frac{1}{2}(x-\theta)^2\right) \exp\left(-\frac{1}{2}\frac{\theta^2}{4}\right).$$

Combining the exponents:

$$p(\theta \mid x) \propto \exp\left[-\frac{1}{2}\left((x-\theta)^2 + \frac{\theta^2}{4}\right)\right].$$

Step 4: Substitute the observed value $x = 5.2$

$$\boxed{p(\theta \mid x = 5.2) \propto \exp\left[-\frac{1}{2}\left((5.2-\theta)^2 + \frac{\theta^2}{4}\right)\right]}$$

(b) Calculate the numerical values of μ_{post} and σ^2_{post} for the given parameters.
Answer: Given:

$$p(x \mid \theta) = \mathcal{N}(x; \theta, \sigma^2), \quad \sigma^2 = 1,$$
$$p(\theta) = \mathcal{N}(\theta; 0, \tau^2), \quad \tau^2 = 4,$$

and the observed value is $x = 5.2$.
Step 1: Posterior form (Gaussian–Gaussian conjugacy)
For a Gaussian likelihood and Gaussian prior, the posterior is also Gaussian:

$$p(\theta \mid x) = \mathcal{N}(\theta; \mu_{\text{post}}, \sigma^2_{\text{post}}),$$

where

$$\sigma^2_{\text{post}} = \left(\frac{1}{\tau^2} + \frac{1}{\sigma^2}\right)^{-1}, \qquad \mu_{\text{post}} = \sigma^2_{\text{post}}\left(\frac{0}{\tau^2} + \frac{x}{\sigma^2}\right).$$

Step 2: Plug in the given parameters

$$\sigma^2_{\text{post}} = \left(\frac{1}{4} + \frac{1}{1}\right)^{-1} = (0.25 + 1)^{-1} = (1.25)^{-1} = 0.8.$$

$$\mu_{\text{post}} = 0.8 \times \left(\frac{0}{4} + \frac{5.2}{1}\right) = 0.8 \times 5.2 = 4.16.$$

$$\boxed{\mu_{\text{post}} = 4.16, \quad \sigma^2_{\text{post}} = 0.8.}$$

Therefore, the posterior distribution is: $p(\theta \mid x = 5.2) = \mathcal{N}(\theta; 4.16, 0.8)$.

(c) Now suppose we do not know the analytical form of the posterior and want to use the *Metropolis–Hastings algorithm* to sample from $p(\theta \mid x)$. Starting at $\theta_t = 2.0$, we propose a new value $\theta' = 3.0$ from a symmetric proposal distribution. Compute the acceptance probability

$$\alpha = \min\left(1, \frac{p(x \mid \theta')\, p(\theta')}{p(x \mid \theta_t)\, p(\theta_t)}\right).$$

Show all steps and give the numerical value of α.

Answer: **Setup:**

$$p(x \mid \theta) = \mathcal{N}(x; \theta, 1) \;\propto\; \exp\!\left(-\tfrac{1}{2}(x-\theta)^2\right), \qquad p(\theta) = \mathcal{N}(\theta; 0, 4) \;\propto\; \exp\!\left(-\tfrac{\theta^2}{8}\right).$$

With a symmetric proposal, the acceptance ratio reduces to the posterior ratio:

$$\alpha \;=\; \min\!\left(1, \frac{p(x \mid \theta')\, p(\theta')}{p(x \mid \theta_t)\, p(\theta_t)}\right) = \min(1,\, r),$$

where

$$\log r \;=\; \left[-\tfrac{1}{2}(x-\theta')^2 - \tfrac{\theta'^2}{8}\right] \;-\; \left[-\tfrac{1}{2}(x-\theta_t)^2 - \tfrac{{\theta_t}^2}{8}\right].$$

Plug in numbers: Let $x = 5.2$, $\theta_t = 2.0$, $\theta' = 3.0$

$$(x-\theta')^2 = (5.2-3.0)^2 = 2.2^2 = 4.84, \qquad (x-\theta_t)^2 = (5.2-2.0)^2 = 3.2^2 = 10.24,$$

$$\frac{\theta'^2}{8} = \frac{9}{8} = 1.125, \qquad \frac{{\theta_t}^2}{8} = \frac{4}{8} = 0.5.$$

Therefore,

$$\log r = \left[-\tfrac{1}{2}\cdot 4.84 - 1.125\right] - \left[-\tfrac{1}{2}\cdot 10.24 - 0.5\right] = (-2.42-1.125)-(-5.12-0.5) = -3.545-(-5.62) = 2.075.$$

Thus

$$r \;=\; e^{2.075} \approx 7.96, \qquad \alpha \;=\; \min(1, r) \;=\; 1.$$

$$\boxed{\alpha = 1}$$